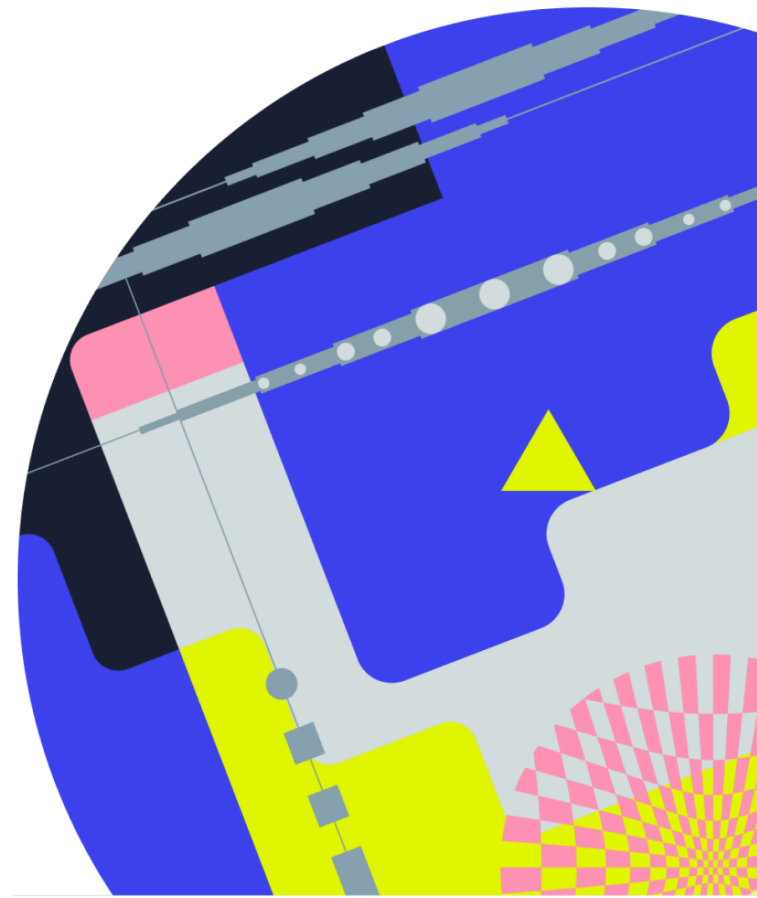


Flow Matching

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Evolution of density (recap)

- $dX_t = f(X_t, t)dt$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t))$$

Continuity
equation

- $dX_t = f(X_t, t)dt + g(t)dW_t$

$$\frac{\partial}{\partial t} p_t(x) = - \frac{\partial}{\partial x} (p_t(x) f(x, t)) + \frac{g^2(t)}{2} \frac{\partial^2}{\partial x^2} p_t(x)$$

Fokker-Planck (-Kolmogorov) equation

Forward and backward (recap)

$$dX_t = f(X_t, t)dt + g(t)dW_t$$

↓

$$dY_t = \left(f(Y_t, t) - \frac{g^2(t)}{2} \nabla \log p_t(Y_t) \right) dt$$

↓

$$dY_t^{(b)} = \left(-f(Y_t^{(b)}, 1-t) + \frac{g^2(1-t)}{2} \nabla \log p_{1-t}(Y_t^{(b)}) \right) dt$$

↓

$$dX_t^{(b)} = \left(-f(X_t^{(b)}, 1-t) + g^2(1-t) \nabla \log p_{1-t}(X_t^{(b)}) \right) dt + g(1-t)dW_t$$

$$\underline{P_{X_t} = P_{Y_t} = P_{Y_{1-t}^{(b)}} = P_{X_{1-t}^{(b)}} \Rightarrow P_{Y_1^{(b)}} = P_{X_1^{(b)}} = P_{\text{data}}}$$

VP/VE-SDE (recap)

$$dX_t = -\frac{\beta_t}{2} X_t dt + \sqrt{\beta_t} dW_t \rightarrow X_t = \sqrt{\alpha_t} X_0 + \sqrt{1-\alpha_t} \varepsilon$$

$$dX_t = g(t) dW_t \rightarrow X_t = X_0 + \sigma_t \varepsilon$$

General case

For each $\alpha_t, \sigma_t \exists f(t), g(t)$ such that

$$dX_t = f(t) X_t dt + g(t) dW_t \rightarrow X_t = \alpha_t X_0 + \sigma_t \varepsilon.$$

PF-ODE (recap)

$$dX_t = f(t)X_t dt + g(t)dW_t$$

↓

$$dY_t = \left(f(t)Y_t - \frac{g^2(t)}{2} \nabla \log p_t(Y_t) \right) dt$$

Sampling: solve $\nearrow$ from $t=1$ to $t=0$.

Denoising Score Matching (recap)

Loss: $X_0 \sim p_{\text{data}}$, $\varepsilon \sim \mathcal{N}(0, I)$, $X_t = \alpha_t X_0 + \beta_t \varepsilon$

$$\int_0^1 \mathbb{E} \left\| S_\theta(X_t, t) - \frac{\partial}{\partial x} \log p_t(X_t | X_0) \right\|^2 dt \rightarrow \min_{\theta}$$

- $p_{t|0}(x_t | x_0) = \mathcal{N}(x_t | \alpha_t x_0, \beta_t^2 I) = C \cdot \exp\left(-\frac{1}{2\beta_t^2} \|x_t - \alpha_t x_0\|^2\right)$

$$\frac{\partial}{\partial x_t} \log p_{t|0}(x_t | x_0) = -\frac{1}{\beta_t^2} (x_t - \alpha_t x_0) = -\frac{\varepsilon}{\beta_t}$$

- Parameterize $S_\theta(x_t, t) = \frac{-\varepsilon_\theta(x_t, t)}{\beta_t}$

Loss: $\int_0^1 \omega_t \mathbb{E} \left\| \varepsilon_\theta(x_t, t) - \varepsilon \right\|^2 dt \rightarrow \min_{\theta}$

Diffusion vs Flow Matching derivation

	Diffusion	Flow Matching
Forward process $dX_t = f(t)X_t dt + g(t)dW_t$	Source of theory	
PF-ODE $dY_t = (f(t)Y_t - \frac{g^2(t)}{2} \frac{\partial}{\partial x} \log p_t(Y_t)) dt$	Learned process	Learned process
Interpolation $X_t = \alpha_t X_0 + \beta_t \epsilon$	Convenient sampling	Source of theory

Diffusion vs Flow Matching derivation

	Diffusion	Flow Matching
Known	Fixed $x_0 \rightarrow$ easy $\nabla \log p_{t 0}(x_t x_0)$	Fixed $x_0, \varepsilon \rightarrow$ easy $x_t = \alpha_t x_0 + \sigma_t \varepsilon$ $\Updownarrow$
Goal	Learn $\nabla \log p_t(x_t)$, sample from PF-ODE $dY_t = (f(t)Y_t - g_2^{(t)} \nabla \log p_t(Y_t)) dt$ that satisfies $p_{Y_t} = p_{x_t}$	$dx_t = v_t(x_t x_0, \varepsilon) dt$ $v_t(x_t x_0, \varepsilon) = \alpha'_t x_0 + \sigma'_t \varepsilon$ Learn $v_t(x)$ s.t. $dY_t = v_t(Y_t) dt$ satisfies $p_{Y_t} = p_{x_t}$

Flow Matching: setting

We know : $dx_t = v_t(x_t | x_0, \varepsilon) dt = (\alpha'_t x_0 + \sigma'_t \varepsilon) dt$

produces dynamics $P_{t|0, \varepsilon}(x_t | x_0, \varepsilon)$.

Goal : find $v_t(x_t)$ s.t. $dY_t = v_t(Y_t) dt$

produces dynamics $P_t(x_t)$.

Key observation :

$dY_t = v_t(Y_t) dt$ produces $P_t \Leftrightarrow$

P_t and v_t satisfy the continuity equation

$$\frac{\partial}{\partial t} P_t(x) = - \frac{\partial}{\partial x} (P_t(x) v_t(x))$$

Flow Matching: derivation

We know: $v_t(x_t | x_0, \varepsilon) = \alpha'_t x_0 + \sigma'_t \varepsilon$ satisfies CE with $p_{t|0, \varepsilon}$.

Goal: find $v_t(x_t)$ that satisfies CE with p_t .

$$\begin{aligned} \frac{\partial}{\partial t} p_t(x) &= \frac{\partial}{\partial t} \int p_{t|0, \varepsilon}(x | x_0, \varepsilon) p_{0, \varepsilon}(x_0, \varepsilon) dx_0 d\varepsilon = \\ &= \int \frac{\partial}{\partial t} p_{t|0, \varepsilon}(x | x_0, \varepsilon) \cdot p_{0, \varepsilon}(x_0, \varepsilon) dx_0 d\varepsilon = \left[p_{t|0, \varepsilon} \text{ sat. CE with } v_t(x | x_0, \varepsilon) \right] \\ &= \int - \frac{\partial}{\partial x} \left(p_{t|0, \varepsilon}(x | x_0, \varepsilon) v_t(x | x_0, \varepsilon) \right) p_{0, \varepsilon}(x_0, \varepsilon) dx_0 d\varepsilon \\ &= - \frac{\partial}{\partial x} \int v_t(x | x_0, \varepsilon) p_{0, \varepsilon, t}(x_0, \varepsilon, t) dx_0 d\varepsilon = \\ &= - \frac{\partial}{\partial x} \left(p_t(x) \underbrace{\int v_t(x | x_0, \varepsilon) p_{0, \varepsilon | t}(x_0, \varepsilon | x) dx_0 d\varepsilon}_{v_t(x)} \right) \end{aligned}$$

Flow Matching : claim

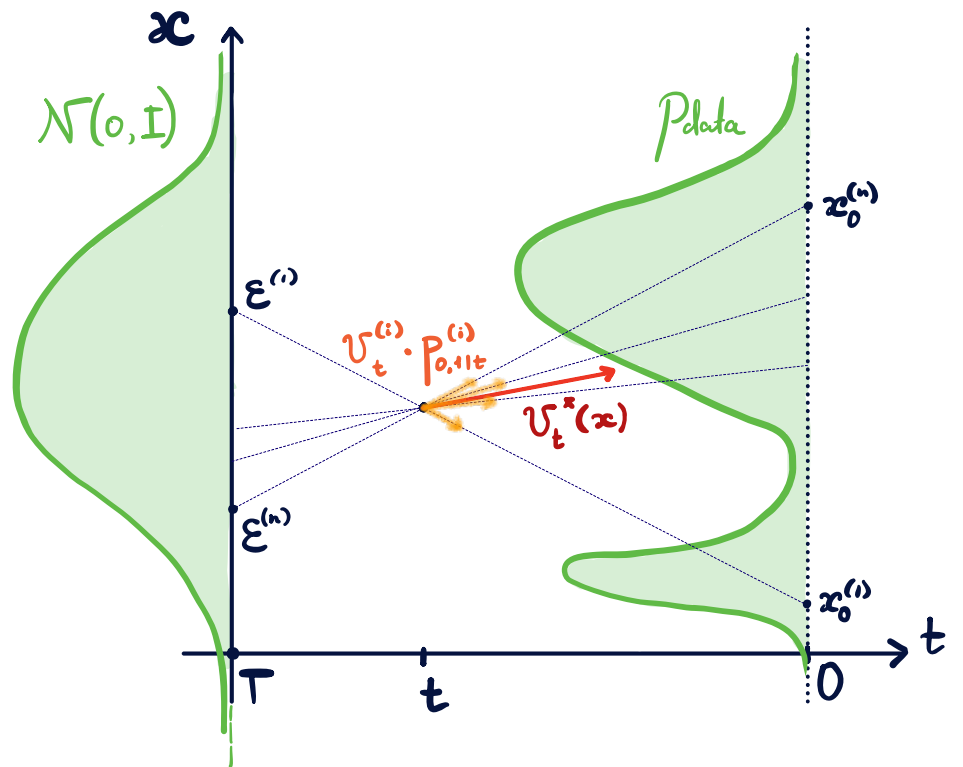
If $v_t(x_t | x_0, \varepsilon) = \alpha'_t x_0 + \sigma'_t \varepsilon$ satisfies CE with $P_{t|0, \varepsilon}$ then

$$v_t^*(x) = \int v_t(x|x_0, \varepsilon) p_{0, \varepsilon|t}(x_0, \varepsilon|x) dx_0 d\varepsilon = \mathbb{E}_{p_{0, \varepsilon|t}(x_0, \varepsilon|x)} v_t(x|x_0, \varepsilon)$$

satisfies CE with p_t .

Interpretation

$v_t^*(x)$: weighted average
of directions from ε to x_0
among pairs (x_0, ε) that
produce $\alpha_t x_0 + G_t \varepsilon = x$



Flow Matching: training

$$v_t^*(x) = \int v_t(x|x_0, \varepsilon) p_{0,\varepsilon|t}(x_0, \varepsilon|x) dx_0 d\varepsilon = \mathbb{E}_{p_{0,\varepsilon|t}(x_0, \varepsilon|x)} v_t(x|x_0, \varepsilon)$$

Same old conditional trick

$$\int_0^T \mathbb{E}_{p_t(x_t)} \|v_t^\theta(x_t) - v_t^*(x_t)\|^2 dt = \int_0^T \mathbb{E}_{p_t(x_t)} \|v_t^\theta(x_t)\|^2 dt + \text{const} \\ - 2 \int_0^T \mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), v_t^*(x_t) \rangle dt =$$

- $\mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), v_t^*(x_t) \rangle = \mathbb{E}_{p_t(x_t)} \langle v_t^\theta(x_t), \mathbb{E}_{p_{0,\varepsilon|t}(x_0, \varepsilon|x_t)} v_t(x_t|x_0, \varepsilon) \rangle = \\ = \mathbb{E}_{p_{0,\varepsilon,t}(x_0, \varepsilon, x_t)} \langle v_t^\theta(x_t), v_t(x_t|x_0, \varepsilon) \rangle.$

$$= \int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0, \varepsilon, x_t)} \|v_t^\theta(x_t) - v_t(x_t|x_0, \varepsilon)\|^2 dt + \text{const}$$

Flow Matching vs Score Matching

$$\int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0, \varepsilon, x_t)} \|\psi_t^\theta(x_t) - \psi_t(x_t | x_0, \varepsilon)\|^2 dt =$$

$$x_t = \alpha_t x_0 + \sigma_t \varepsilon, \quad \psi_t(x_t | x_0, \varepsilon) = \alpha_t' x_0 + \sigma_t' \varepsilon.$$

$$= \int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0, \varepsilon, x_t)} \|\psi_t^\theta(x_t) - (\alpha_t' x_0 + \sigma_t' \varepsilon)\|^2 dt =: \mathcal{L}_{\text{FM}}$$

$$x_0 = \frac{x_t - \sigma_t \varepsilon}{\alpha_t} \quad \varepsilon = \frac{x_t - \alpha_t x_0}{\sigma_t}$$

$$\psi_t^\theta(x_t) = \alpha_t' \frac{x_t - \sigma_t \varepsilon_t^\theta(x_t)}{\alpha_t} + \sigma_t' \varepsilon_t^\theta(x_t) \Rightarrow \mathcal{L}_{\text{FM}} = \int_0^T \mathbb{E}[A_t^2 \|\epsilon_t^\theta(X_t) - \epsilon\|^2] dt$$

$$\psi_t^\theta(x_t) = \alpha_t' x_0^\theta(x_t) + \sigma_t' \frac{x_t - \alpha_t x_0^\theta(x_t)}{\sigma_t} \Rightarrow \mathcal{L}_{\text{FM}} = \int_0^T \mathbb{E}[B_t^2 \|X_0^\theta(X_t) - X_0\|^2] dt$$

$$A_t = \sigma_t' - \frac{\alpha_t' \sigma_t}{\alpha_t}, \quad B_t = \alpha_t' - \frac{\sigma_t' \alpha_t}{\sigma_t}$$

$$\alpha_t = 1 - t, \quad \sigma_t = t: \quad A_t^2 = \frac{1}{(1-t)^2}, \quad B_t^2 = \frac{1}{t^2} \quad (0 < t < 1).$$

Flow Matching vs diffusion models

Training: $\int_0^T \mathbb{E}_{p_{0,\varepsilon,t}(x_0, \varepsilon, x_t)} \|v_t^\theta(x_t) - v_t(x_t|x_0, \varepsilon)\|^2 dt$

$\Leftrightarrow$ diffusion

Sampling: $dY_t = v_t^\theta(Y_t)dt$ vs $dY_t = (f(t)Y_t - \frac{g^2(t)}{2} s_t^\theta(Y_t))dt$?

Claim Theoretical optimum $v_t^*(x)$ is equal to

$f(t)x - \frac{g^2(t)}{2} \nabla \log p_t(x) \Leftrightarrow$ diffusion PF-ODE

Conclusion Flow Matching $\equiv$ Diffusion

- Only in case $p_0(x) = \mathcal{N}(x|\dots)$ and $x_t = \alpha_t x_0 + \beta_t \varepsilon$
- Extends to image-to-image
- Different point of view: $\alpha_t x_0 + \beta_t \varepsilon$ is simpler than SDEs.

Linear interpolant and straight trajectories

- $X_t = t\varepsilon + (1-t)X_0 \Rightarrow v_t(X_t | X_0, \varepsilon) = \varepsilon - X_0$
straight conditional trajectories (motivation behind FM)
straighter $\Rightarrow$ fewer points

- Are trajectories of $dY_t = v_t^*(Y_t)dt$ straight?

- Let $p_{\text{data}}(x_0) = \mathcal{N}(x_0 | 0, I)$.

Then $X_t = t\varepsilon + (1-t)X_0 \sim \mathcal{N}(0, (t^2 + (1-t)^2)I)$.

If Y_t is straight, then $p_{Y_0} = p_{Y_1} = \mathcal{N}(0, I) \Rightarrow p_{Y_t} = \mathcal{N}(0, I)$.

But $p_{Y_t} = p_{X_t} = \mathcal{N}(0, (t^2 + (1-t)^2)I)$.

Generally, no

- More separated and concentrated modes
 $\Rightarrow$ straighter trajectories.